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Proper conflict-free 6-coloring of planar graphs without short cycles ☆

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A B S T R A C T

A proper conflict-free l-coloring of a graph G is a proper l-coloring satisfying that for any non-isolated vertex $v \in V(G)$, there exists a color appearing exactly once in $N_G(v)$. The proper conflict-free chromatic number, denoted by $\chi_{pcf}(G)$, is the minimal integer l so that G admits a proper conflict-free l-coloring. This notion was proposed by Fabrici et al. in 2022. They focus mainly on proper conflict-free coloring of outerplanar graphs and planar graphs. They constructed a planar graph that has no proper conflict-free 5-coloring and conjectured every planar graph G has $\chi_{pcf}(G) \le 6$. In this paper, we confirm this conjecture for planar graphs without cycles of lengths 3, 5 or 6.

1. Introduction

Motivated by the frequency assignment problem, Even et al. [6] introduced the hypergraph's conflict-free coloring. A conflict-free coloring of a hypergraph $(\mathcal{V}, \mathcal{E})$ is a coloring of each vertex in $\mathcal{V}$ so that in each $e \in \mathcal{E}$, there is at least one color appearing only on one vertex in e.

We pay attention to the proper conflict-free coloring of general graphs. A *proper l-coloring* of a graph G is a mapping $g : V(G) \to [n]$ such that $g(u) \ne g(v)$ whenever $uv \in E(G)$, where $[n] = \{1, 2, \dots, n\}$. A *proper conflict-free l-coloring (PCF l-coloring for short)* of G is a strengthened version of proper l-coloring, which requires for any non-isolated vertex $v \in V(G)$, the color set of $N_G(v)$ has a color occurring exactly once. The *proper conflict-free chromatic number*, denoted by $\chi_{pcf}(G)$, is the smallest integer l so that G admits a PCF l-coloring. The *square* of a graph G is obtained from G by adding all the edges between vertices u and v with $d_G(u, v) \le 2$. The PCF l-coloring is an intermediate concept between proper l-coloring of a graph and proper l-coloring of a graph's square. This notion was formally introduced by Fabrici et al. [7] in 2022. They proved that every planar graph has a PCF 8-coloring and constructed a planar graph that has no PCF 5-coloring. They gave the following conjecture.

Conjecture 1.1. *For every planar graph G, $\chi_{pcf}(G) \le 6$.*

Recently, proper conflict-free coloring has already attracted much attention [1,3,4,8–10]. Caro, Petruševski and Škrekovski [5] determined the $\chi_{pcf}(G)$ for some fundamental graph classes such as trees, hypercubes, cycles and subdivisions of complete graphs. They also studied the upper bound on the maximum average degree of graphs that guarantee a PCF l-coloring for $l \in \{4, 5, 6\}$. Liu, Reed [11] showed that every graph has a proper conflict-free coloring with $(1 + o(1))\Delta$ colors.

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Cho, Choi, Kwon and Park [2] proved that for $l \geq 5$, a graph G with $mad(G) \leq \frac{4l}{l+2}$ has a PCF l-coloring, unless G has a 1-subdivision of the complete graph on $l+1$ vertices. This result implies that $\chi_{pcf}(G) \leq 6$ for every planar graph G without cycles of lengths 3, 4 or 5. In this paper, we confirm Conjecure 1.1 for planar graphs without cycles of lengths 3, 5 or 6.

Theorem 1.2. *For every planar graph G without cycles of lengths 3, 5 or 6, $\chi_{pcf}(G) \leq 6$.*

Here are some notation. For a vertex u, we call u an *l-vertex*, or *l^+-vertex*, or *l^--vertex* if $d(u) = l$, or $d(u) \geq l$, or $d(u) \leq l$, respectively. The same notation will be applied to faces and cycles. An l-face $f = [u_1u_2 \cdots u_l]$ is an *$(x_1, x_2 \ldots, x_l)$-face* if $d(u_i) = x_i$ for each $i \in [l]$. Let w be a neighbor of a vertex x in graph G. We call w an l-neighbor or *l^+-neighbor*, or *l^--neighbor* of x if $d(w) = l$, or $d(w) \geq l$, or $d(w) \leq l$, respectively. Let $N_l(x)$ be the set of all l-neighbors of x in G and $n_l(x) = |N_l(x)|$. Similarly, we define $N_{l+}(x)$. A conflict-free color of x is a color appearing exactly once on $N_G(x)$. If Γ is a PCF coloring of G, we use $\Gamma_*(x)$ to denote the conflict-free color of x, if it exists. (we will choose one arbitrarily if there are two or more conflict-free colors of x.) Regard $\Gamma_*(x)$ as automatically replaced when the coloring of the graph changes.

2. Features of minimal counterexample

Suppose G is a planar graph without cycles of lengths 3, 5 or 6 that has no PCF 6-coloring, but each planar graph $\widetilde{G}$ without cycles of lengths 3, 5 or 6 such that $|V(\widetilde{G})| < |V(G)|$ is PCF 6-colorable, and fix a plane embedding of G. Obviously G is a minimal counterexample to Theorem 1.2. Let us investigate the structural properties of G.

Lemma 2.1. $\delta(G) \geq 2$.

Proof. Suppose otherwise that $d(x) = 1$ and $xy \in E(G)$. $G - x$ has a PCF 6-coloring Γ by the choice of G. We color x by avoiding $\Gamma(y)$ and $\Gamma_*(y)$. Note that $\Gamma(y)$ is the conflict-free color of x. So G has a PCF 6-coloring, a contradiction. □

Lemma 2.2. *If x and y are two 3^--vertices in G, then $xy \notin E(G)$.*

Proof. Suppose otherwise that $xy \in E(G)$. By Lemma 2.1, $d(x), d(y) \geq 2$. We consider the case that $d(x) = d(y) = 3$, the other cases are easier. Let $N(x) = \{x_1, x_2, y\}$, $N(y) = \{y_1, y_2, x\}$. $G \setminus \{x, y\}$ has a PCF 6-coloring Γ by the choice of G. Then we color x with c_1 by avoiding at most five colors in $\{\Gamma(x_1), \Gamma(x_2), \Gamma_*(x_1), \Gamma_*(x_2), \Gamma(y_1)\}$. If $\Gamma(x_1) \neq \Gamma(x_2)$, then y can be colored by avoiding at most five colors in $\{c_1, \Gamma(y_1), \Gamma_*(y_1), \Gamma(y_2), \Gamma_*(y_2)\}$. Note that at least one of $\{\Gamma(x_1), \Gamma(x_2)\}$ is the conflict-free color of x, at least one of $\{c_1, \Gamma(y_1)\}$ is the conflict-free color of y. We get a PCF 6-coloring of G, a contradiction. If $\Gamma(x_1) = \Gamma(x_2)$, then we color y with c_2 by avoiding at most five colors in $\{\Gamma(x_1), \Gamma(y_1), \Gamma_*(y_1), \Gamma(y_2), \Gamma_*(y_2)\}$ and recolor x with a color c_3 by avoiding at most five colors in $\{\Gamma(x_1), \Gamma_*(x_1), \Gamma_*(x_2), \Gamma(y), \Gamma(y_1)\}$. Note that c_2 is the conflict-free color of x, at least one of $\{c_3, \Gamma(y_1)\}$ is the conflict-free color of y. Thus G has a PCF 6-coloring, also a contradiction. □

Lemma 2.3. *The graph G contains no $(2, 4^+, 2, 4^+)$-faces. Consequently, each 4-face is incident to at most one 2-vertex.*

Proof. Suppose otherwise that $f = [uvwx]$ is a $(2, 4^+, 2, 4^+)$-face in G. $G - u$ has a PCF 6-coloring Γ by the choice of G. Then we color u by avoiding at most four colors in $\{\Gamma(v), \Gamma_*(v), \Gamma(x), \Gamma_*(x)\}$. Note that $\Gamma(v) \neq \Gamma(x)$ since $d(w) = 2$. So $\Gamma(v)$ or $\Gamma(x)$ is the conflict-free color of u. We get a PCF 6-coloring of G, a contradiction. Since G contains no adjacent 2-vertices by Lemma 2.2, there is at most one 2-vertex on f. □

For convenience, a vertex x is *bad* in G if x satisfies one of the following statements: (i) $d(x) = 4$ and x has exactly two 2-neighbors; (ii) $d(x) = 5$ and x has exactly four 2-neighbors. A vertex x is *semi-bad* if x satisfies one of the following statements: (i) $d(x) = 4$ and x has exactly one 2-neighbor; (ii) $d(x) = 5$ and x has exactly three 2-neighbors. A *k_t-vertex* is a k-vertex with t 2-neighbors. A vertex x is a *special 5_2-vertex* in G if x has two 2-neighbors and x is on three 4-faces. In this case, let $N(x) = \{x_i : 1 \leq i \leq 5\}$ and g_i be the face containing x_i, x, x_{i+1} (index modulo 5). Since G has no 6-cycles, two adjacent 4-faces must share a 2-vertex. By Lemma 2.3 G doesn't have $(2, 4^+, 2, 4^+)$-faces. So by symmetry we have g_1, g_2, g_4 are 4-faces and $d(x_2) = d(x_4) = 2$. We call a vertex *good* if it is a 4^+-vertex but not bad or semibad or a special 5_2-vertex.

Lemma 2.4. *Let $f = [uvwx]$ be a 4-face in G with $d(u) = 2$. Then the following cases do not hold:*

(1) *v is semi-bad, w is a 5_1-vertex and x is bad;*
(2) *v, w are semi-bad, x is a special 5_2-vertex;*
(3) *w is semi-bad, v, x are 5-vertices;*
(4) *v is semi-bad and $d(w) = 3$;*
(5) *v is a 4- or 5_2-vertex, x is bad and w is a 4- or 5_2-vertex.*

Proof. (1) Let v_1, v_2 be adjacent to v not on f if $d(v) = 4$ and v_1, v_2, v_3 be adjacent to v not on f with $d(v_1) = d(v_2) = 2$ if $d(v) = 5$. Let w_1, w_2, w_3 be adjacent to w not on f with $d(w_1) = 2$. Let x_1, x_2 be adjacent to x not on f with $d(x_1) = 2$ if $d(x) = 4$ and x_1, x_2, x_3 be adjacent to x not on f with $d(x_1) = d(x_2) = d(x_3 = 2)$ if $d(x) = 5$. Let $S = V(f) \cup N_2(v) \cup N_2(w) \cup N_2(x)$. $G - S$ has a PCF 6-coloring Γ by the choice of G. We color w with a color c_1 by avoiding at most five colors in $\{\Gamma_*(w_1), \Gamma_*(w_2), \Gamma_*(w_3), \Gamma(w_2), \Gamma(w_3)\}$. Color v with a color c_2 by avoiding at most five colors in $\{c_1, \Gamma_*(v_1), \Gamma_*(v_2), \Gamma(v_1), \Gamma(v_2)\}$ if $d(v) = 4$ and colors in $\{c_1, \Gamma_*(v_1), \Gamma_*(v_2), \Gamma_*(v_3), \Gamma(v_3)\}$ if $d(v) = 5$. We color x by avoiding colors in $\{c_1, c_2, \Gamma_*(x_1), \Gamma(x_2), \Gamma_*(x_2)\}$ if $d(v) = 4$ and colors in $\{c_1, c_2, \Gamma_*(x_1), \Gamma_*(x_2), \Gamma_*(x_3)\}$ if $d(v) = 5$.

For each 2-vertex $a \in S$, we may suppose that $N(a) = \{a_1, a_2\}$, and $\Gamma_*(a_1)$ is not defined. Then color a with c by avoiding colors in $\{\Gamma(a_1), \Gamma(a_2), \Gamma_*(a_2)\}$, and avoiding at most $\lfloor \frac{4}{2} \rfloor = 2$ colors in the color set of $N_G(a_1)$ since $d(a_1) \le 5$. Now a_1 has a conflict-free color c. We extend Γ to G, a contradiction.

(2) Let v_1, v_2 be adjacent to v not on f if $d(v) = 4$ and v_1, v_2, v_3 be adjacent to v not on f with $d(v_1) = d(v_2) = 2$ if $d(v) = 5$. Let w_1, w_2 be adjacent to w not on f with $d(w_1) = 2$ if $d(w) = 4$ and w_1, w_2, w_3 be adjacent to w not on f with $d(w_1) = d(w_2) = d(w_3) = 2$ if $d(w) = 5$. Let x_1, x_2, x_3 be adjacent to x not on f with $d(x_1) = 2$. Let $S = V(f) \cup N_2(w) \cup N_2(x) \cup N_2(v)$. $G - S$ has a PCF 6-coloring Γ by the choice of G. We apply a color c_1 to x by avoiding colors in $\{\Gamma_*(x_1), \Gamma(x_2), \Gamma(x_3), \Gamma_*(x_2), \Gamma_*(x_3)\}$. We apply a color c_2 to v by avoiding colors in $\{c_1, \Gamma(v_1), \Gamma(v_2), \Gamma_*(v_1), \Gamma_*(v_2)\}$ if $d(v) = 4$ and colors in $\{c_1, \Gamma_*(v_1), \Gamma_*(v_2), \Gamma(v_3), \Gamma_*(v_3)\}$ if $d(v) = 5$. We color w by avoiding colors in $\{c_1, c_2, \Gamma_*(w_1), \Gamma(w_2), \Gamma_*(w_2)\}$ if $d(w) = 4$ and colors in $\{c_1, c_2, \Gamma_*(w_1), \Gamma_*(w_2), \Gamma_*(w_3)\}$ if $d(w) = 5$. Similar to above, we color each 2-vertex in S by avoiding at most five colors. Now we extend Γ to G, a contradiction.

(3) Let w_1, w_2 be adjacent to w not on f with $d(w_1) = 2$ if $d(w) = 4$ and w_1, w_2, w_3 be adjacent to w not on f with $d(w_1) = d(w_2) = d(w_3) = 2$ if $d(w) = 5$. Let $S = \{w\} \cup N_2(w)$. By the choice of G, $G - S$ has a PCF 6-coloring Γ. We color w by avoiding colors in $\{\Gamma_*(w_1), \Gamma(w_2), \Gamma_*(w_2), \Gamma(v), \Gamma(x)\}$ if $d(w) = 4$ and colors in $\{\Gamma_*(w_1), \Gamma_*(w_3), \Gamma_*(w_2), \Gamma(v), \Gamma(x)\}$ if $d(w) = 5$. Similar to above, we can color each 2-vertex in S by avoiding at most five colors. We will show that u can be recolored by avoiding at most five colors to guarantee the existence of conflict-free colors of v and x. If none of $\{v, x\}$ has a conflct-free color, then each of $N_G(v)$ and $N_G(x)$ has at most two colors. Since w is the common neighbor of v and x, we can recolor u by avoiding Γ(v),Γ(x) and at most $2 + 2 - 1 = 3$ colors in the color set of $N_G(v) \cup N_G(x)$. If one of $\{v, x\}$ has a conflct-free color, say v, we can recolor u by avoiding $\{\Gamma(v), \Gamma(x), \Gamma_*(v)$ and at most two colors in the color set of $N_G(x)$. Now we get a PCF 6-coloring of G_1, a contradiction.

(4) Let v_1, v_2 be adjacent to v not on f if $d(v) = 4$ and v_1, v_2, v_3 be adjacent to v not on f with $d(v_1) = d(v_2) = 2$ if $d(v) = 5$. Let w_1 be the neighbor of w not on f. Let $S = \{u, v, w\} \cup N_2(v)$. By the choice of G, $G - S$ has a PCF 6-coloring Γ. We color v with c by avoiding colors in $\{\Gamma(x), \Gamma(v_1), \Gamma_*(v_1), \Gamma(v_2), \Gamma_*(v_2)\}$ if $d(v) = 4$ and colors in $\{\Gamma(x), \Gamma_*(v_1), \Gamma_*(v_2), \Gamma(v_3), \Gamma_*(v_3)\}$ if $d(v) = 5$. Then we color w by avoiding colors in $\{\Gamma(w_1), \Gamma_*(w_1), c, \Gamma(x), \Gamma_*(x)\}$, color u by avoiding colors in $\{c, \Gamma(x), \Gamma_*(x)\}$. Similar to above, we color each 2-neighbor of v by avoiding at most five colors. Now we extend Γ to G, a contradiction.

(5) Let v_1, v_2 be adjacent to v not on f if $d(v) = 4$ and v_1, v_2, v_3 be adjacent to v not on f with $d(v_1) = 2$ if $d(v) = 5$. Let w_1, w_2 be adjacent to w not on f if $d(w) = 4$ and w_1, w_2, w_3 be adjacent to w not on f with $d(w_1) = d(w_2) = 2$ if $d(w) = 5$. Let x_1, x_2 be adjacent to x not on f with $d(x_1) = 2$ if $d(x) = 4$ and x_1, x_2, x_3 be adjacent to x not on f with $d(x_1) = d(x_2) = d(x_3) = 2$ if $d(x) = 5$. Let $S = V(f) \cup N_2(v) \cup N_2(w) \cup N_2(x)$. $G - S$ has a PCF 6-coloring Γ by the choice of G. We color v with c_1 by avoiding colors in $\{\Gamma_*(v_1), \Gamma_*(v_2), \Gamma(v_1), \Gamma(v_2)\}$ if $d(v) = 4$ and colors in $\{\Gamma_*(v_1), \Gamma(v_2), \Gamma_*(v_2), \Gamma(v_3), \Gamma_*(v_3)\}$ if $d(v) = 5$. We color w with c_2 by avoiding colors in $\{c_1, \Gamma(w_1), \Gamma(w_2), \Gamma_*(w_1), \Gamma_*(w_2)\}$ if $d(v) = 4$ and colors in $\{\Gamma_*(w_1), \Gamma_*(w_2), c_1, \Gamma(w_3), \Gamma_*(w_3)\}$ if $d(v) = 5$. Color x by avoiding colors in $\{c_1, c_2, \Gamma_*(x_1), \Gamma(x_2), \Gamma_*(x_2)\}$ if $d(x) = 4$ and colors in $\{c_1, c_2, \Gamma_*(x_3), \Gamma_*(x_2), \Gamma_*(x_1)\}$ if $d(x) = 5$. Similar to above, we can color each 2-vertex in $N_2(v) \cup N_2(w) \cup N_2(x)$ by avoiding at most five colors. Now G has a PCF 6-coloring Γ, a contradiction. □

Lemma 2.5. *Let x be a vertex in G with at least one 2-neighbor x' and $d(x) \in \{4, 5, 6\}$. Let $L = \{x_1, x_2, ..., x_l\}$ be the set of bad neighbors of x, let $K = \{x_{l+1}, x_{l+2}, ..., x_{l+k}\}$ be the set of semi-bad neighbors of x, then $2d(x) - 2l - k - n_2(x) - n_3(x) \ge 6$.*

Proof. Suppose otherwise that $2d(x) - 2l - k - n_2(x) - n_3(x) \le 5$. Let $S = \{x\} \cup N_2(x) \cup N_3(x) \cup (\cup_{1 \le i \le l+k} N_2(x_i)) \cup L \cup K$. $G - S$ has a PCF 6-coloring Γ by the choice of G. Let $N_i'(x)$ be the set of vertices consisting of a neighbor y for each vertex in $N_i(x)$ where $y \ne x$, $i \in \{2, 3\}$. For each vertex $z \in K$, let $C_1 = \Gamma(N_{3+}(z) \cup N_2{}'(z)) \cup \Gamma_*(N_{3+}(z))$. Since z is semi-bad, $|C_1| \le 5$. So we can extend Γ to z by avoidng colors in C_1. Next we apply a color to x that does not appear in $C_2 = \Gamma(N_{4+}(x) \cup N_2{}'(x) \cup N_3{}'(x)) \cup \Gamma_*(N_{4+}(x) \backslash (K \cup L))$, which is possible since $|C_2| = (d(x) - l) + (d(x) - n_2(x) - n_3(x) - k - l) = 2d(x) - 2l - k - n_2(x) - n_3(x) \le 5$ by our assumption. Color $w \in N_3(x)$ with a color that does not appear in $\Gamma(N(w)) \cup \Gamma_*(N(w) \backslash \{x\})$, which is possible since $|\Gamma(N(w)) \cup \Gamma_*(N(w) \backslash \{x\})| \le 5$. Color each vertex $y \in L$ with a color not in $C_3 = \Gamma(N_{3+}(y) \cup N_2{}'(y)) \cup \Gamma_*(N_{3+}(y) \backslash \{x\})$, which is possible since $|C_3| \le 4 + |\{\Gamma(x)\}| = 5$. Now we can color each 2-vertex in S by avoiding at most $3 + \lfloor \frac{5}{2} \rfloor = 5$ colors to guarantee the existence of the conflict-free colors of vertices in $K \cup L \cup \{x\}$. So G has a PCF 6-coloring, a contradiction. □

Corollary 2.6. *Each statement of the following holds in G:*

(1) *Let $xy \in E(G)$. If x is bad, then y cannot be bad, semi-bad or a 3-vertex;*
(2) *Each semi-bad vertex is incident to at most one semi-bad vertex;*
(3) *A 4-vertex in G has at most two 2-neighbors and a 5-vertex in G has at most four 2-neighbors;*
(4) *A semi-bad vertex in G has at most one 3-neighbor.*

Proof. (1) Suppose otherwise that y is bad, semi-bad or a 3-vertex. Since x is bad, either x is a 5-vertex with four 2-neighbors or a 4-vertex with two 2-neighbors. Then $2d(y) - 2l - k - n_2(y) - n_3(y) \le 2d(y) - n_2(y) - 1 \le 5$, contrary to Lemma 2.5.

(2) Suppose otherwise that a semi-bad vertex x is incident to at least two semi-bad vertices x_1, x_2. Then $2d(y) - 2l - k - n_2(y) - n_3(y) \le 2d(y) - n_2(y) - 2 \le 5$, contrary to Lemma 2.5.

(3) By Lemma 2.5, $2d(y) - n_2(y) \ge 2d(y) - 2l - k - n_2(y) - n_3(y) \ge 6$, so $n_2(y) \le 2d(y) - 6 = 2$ if $d(y) = 4$ and $n_2(y) \le 2d(y) - 6 = 4$ if $d(y) = 5$. So a 4-vertex in G has at most two 2-neighbors, a 5-vertex in G has at most four 2-neighbors.

(4) By Lemma 2.5, $2d(v) - n_2(v) - n_3(v) \ge 2d(v) - 2l - k - n_2(v) - n_3(v) \ge 6$, so $n_3(v) \le 2d(v) - n_2(v) - 6 = 1$ if v is semi-bad. So a semi-bad vertex in G has at most one 3-neighbor. □

A 4-face $g = [uvwx]$ is *poor* if g satisfies one of the followings:(I) $d(u) = 2$, w is bad or a 3-vertex, x is a special 5_2-vertex; (II) $d(u) = 2$, v is a special 5_2-vertex, w is a 5_1-vertex and x is bad. Furthermore, a 4-face is I-poor if it satisfies (I) and II-poor if it satisfies (II).

Lemma 2.7. *Let x be a special 5_2-vertex in G. Then x is incident to at most one poor 4-face.*

Proof. Let $N(x) = \{x_1, x_2, x_3, x_4, x_5\}$ and g_i be the face containing x_i, x, x_{i+1} (index module 5). By the definition of special 5_2-vertices, g_1, g_2, g_4 are 4-faces and $d(x_2) = d(x_4) = 2$. First we claim that if x is incident to a I-poor face, say g_1 (in this case, x_1 is bad or a 3-vertex), then x is incident to no bad vertices or 3-vertices not on g_1. Without lose of generality, suppose otherwise that x_3 is bad or a 3-vertex. Let $S = \{x_1, x_2, x_3, x_4, x\} \cup N_2(x_1) \cup N_2(x_3)$. By the choice of G, $G - S$ has a PCF 6-coloring Γ. Color x by avoiding colors in $\{\Gamma_*(x_2), \Gamma_*(x_4), \Gamma(x_5), \Gamma_*(x_5)\}$ and $\Gamma_*(x_3)$ if $d(x_3) = 3$. Then we color each of x_1 and x_3 by avoiding at most five colors. Then we color each 2-vertex in S by avoiding at most five colors. We get a PCF 6-coloring of G, a contradiction.

By the above claim x is incident to at most one I-poor face. So we consider the following four cases.

Case 1: g_1 is a I-poor face and g_2 is a II-poor face. Let $g_1 = [xx_1yx_2]$. Then y is bad, x_1 is bad or 3-vertex. We get a bad vertex incident to a bad or 3-vertex, contrary to Corollary 2.6(1).

Case 2: g_1 is a II-poor face and g_2 is a II-poor face. In this case, x_1 and x_3 are 5_1-vertices. Let y be the bad vertex adjacent to x_2. Note that $d(y) = 4$. Let y' be the neighbor of y not on g_1 or g_2. Let $S = \{x, x_1, x_2, x_3, x_4, y\} \cup N_2(y) \cup N_2(x_1) \cup N_2(x_3)$. By the choice of G, $G - S$ has a PCF 6-coloring Γ. Color each of x_1, x_3 with c_1, c_2 by avoiding at most five colors,respectively. Then color x with c_3 by avoiding colors in $\{c_1, c_2, \Gamma_*(x_4), \Gamma(x_5), \Gamma_*(x_5)\}$. Then color y by avoiding at most four colors in $\{c_1, c_2, c_3, \Gamma_*(y')\}$. Then color each 2-vertex in S by avoiding at most five colors. Now G has a PCF 6-coloring, a contradiction.

Case 3: g_1 is a I-poor face and g_4 is a II-poor face. In this case, x_1 is bad or 3-vertex and x_5 is a 5_1-vertex. Let y be the bad vertex adjacent to x_4. Let $S = \{x, x_1, x_2, x_4, x_5, y\} \cup N_2(y) \cup N_2(x_1) \cup N_2(x_5)$. By the choice of G, $G - S$ has a PCF 6-coloring Γ. Color x_5 with c by avoiding at most five colors. Then color y by avoiding at most four colors. Color x by avoiding colors in $\{\Gamma_*(x_2), \Gamma_*(x_3), \Gamma(x_3), c, \Gamma_*(x_4)\}$. Then color x_1 by avoiding at most five colors. Then color each 2-vertex in S by avoiding at most five colors. Now G has a PCF 6-coloring, a contradiction.

Case 4: g_1 is a II-poor face and g_4 is a II-poor face. In this case, x_1 and x_5 are 5_1-vertices. Let y_1, y_2 be the bad vertex adjacent to x_2, x_4, respectively. Let $S = \{x, x_1, x_2, x_4, x_5, y_1, y_2\} \cup N_2(y_1) \cup N_2(y_2) \cup N_2(x_1) \cup N_2(x_5)$. By the choice of G, $G - S$ has a PCF 6-coloring Γ. Color each of x_1, x_5 with c_1, c_2 by avoiding at most five colors, respectively. Then color x by avoiding colors in $\{c_1, c_2, \Gamma(x_3), \Gamma_*(x_3)\}$. Then color each of y_1, y_2 by avoiding at most five colors. Then color each 2-vertex in S by avoiding at most five colors. Now G has a PCF 6-coloring, a contradiction. □

3. Discharging

In this part, we will finish the proof of Theorem 1.2 by discharging method on the minimal counterexample G. For every vertex $x \in V(G)$, let its initial charge be $\eta(x) = 2d(x) - 6$, for each face $g \in F(G)$, let its initial charge be $\eta(g) = d(g) - 6$. Using Euler's formula, $\sum_{x \in V(G)} \eta(x) + \sum_{g \in F(G)} \eta(g) = -12$. We will make a set of rules to redistribute the charge among faces and vertices of G, so that every face and vertex has non-negative final charge. It would give us a contradiction because of $-12 = \sum_x \eta(x) + \sum_g \eta(g) = \sum_x \eta^*(x) + \sum_g \eta^*(g) \ge 0$.

The discharging rules are as follows.

(R1) Each 4^+-vertex sends 1 to each incident 2-vertex.
(R2) Each semi-bad vertex sends $\frac{1}{2}$ to each incident 4-face.
(R3) Each special 5_2-vertex sends 1 to each incident poor 4-face and $\frac{1}{2}$ to each other incident 4-face.
(R4) Let v be a good 4- or 5-vertex incident to a 4-face f in G. If v is a 4_0-vertex or a 5_1-vertex or a non-special 5_2-vertex, then v sends 1 to f; if v is a 5_0-vertex, then v sends 2 to f.
(R5) Each 6^+-vertex sends 1 to each incident $(2, 6^+, 3^+, 4^+)$-face and 2 to each incident $(3^+, 6^+, 3^+, 2^+)$-face.

Lemma 3.1. *Each face g in G has $\eta^*(g) \ge 0$.*

Proof. Since G contains no 3- 5- or 6-cycles, $d(g) = 4$ or $d(g) \ge 7$. Note that 7^+-faces are not involved in the rules. Thus, every 7^+-face has $\eta^*(g) \ge 1$. We only need to check 4-faces. Let $g = [uvwx]$ be a 4-face in G. We are going to consider the following four cases.

Case 1: g has no 3^--vertices or bad vertices. By the definition of bad vertices, each vertex on g is a 4-vertex with at most one incident 2-vertex, or a 5-vertex with at most three incident 2-vertices, or a 6^+-vertex. By (R2)-(R5) g gets at least $\frac{1}{2}$ from each incident vertex. So $\eta^*(g) \geq -2 + \frac{1}{2} \times 4 = 0$.

Case 2: g is incident to some bad vertex, say u, but no 3^--vertices. Then g is a $(4^+, 4^+, 4^+, 4^+)$-face. Observe that each vertex on g cannot be a special 5_2-vertex. By Corollary 2.6(1) v and x are good 4^+-vertices. By (R2)-(R5) g gets at least 1 from each of v and x. So $\eta^*(g) \geq -2 + 1 + 1 = 0$.

Case 3: g is incident to some 3^--vertices, but no bad vertices. By Lemma 2.2 g is incident to at most two 3^--vertices.

Case 3.1: g is incident to exactly one 3^--vertex, say u. Then we will show that at least one of v, w, x is good. Suppose otherwise, then each of v, w, x is either semi-bad or a special 5_2-vertex. By Corollary 2.6(2) at most two of v, w, x are semi-bad. We have the following two cases:(1) v, w are semi-bad, x is a special 5_2-vertex. This is contrary to Lemma 2.4(2). (2)w is semi-bad, v, x are special 5_2-vertices. This is contrary to Lemma 2.4(3). So we may assume that v is good. By (R2)-(R5) g gets at least 1 from v and at least $\frac{1}{2}$ from each of w and x. So $\eta^*(g) \geq -2 + 1 + \frac{1}{2} \times 2 = 0$.

Case 3.2: g is incident to exactly two 3^--vertices. Then g must be a $(2, 4^+, 3, 4^+)$-face or a $(3, 4^+, 3, 4^+)$-face by Lemma 2.2 and Lemma 2.3. If f is a $(2, 4^+, 3, 4^+)$-face, then each of v and x cannot be semi-bad by Lemma 2.4(4). So each of v and x is good or a special 5_2-vertex. If at least one of v, x is a special 5_2-vertex, then g is a I-poor face. Otherwise, both v and x are good. In either case, g gets 1 from each of v and x by (R3)-(R5). So $\eta^*(g) \geq -2 + 1 + 1 = 0$. In the latter case, by Corollary 2.6(4) and the definition of special 5_2-vertex, each of v and x cannot be semi-bad or a special 5_2-vertex. So v and x are good. By (R2)-(R5) g gets at least 1 from each of v and x. So $\eta^*(g) \geq -2 + 1 + 1 = 0$. If g is a $(3, 4^+, 3, 4^+)$-face, then none of v, x is semi-bad or a special 5_2-vertex by Corollary 2.6(4) and the definition of special 5_2-vertices. So v, x are good. By (R4)(R5) g gets at least 1 from each of v and x. So $\eta^*(g) \geq -2 + 1 + 1 = 0$.

Case 4: g is incident to both bad vertices and 3^--vertices. By symmetry we consider the following two cases.

Case 4.1: u is a 3^--vertex and w is bad. By Corollary 2.6(1) each of v and x cannot be bad or semi-bad. If $d(u) = 3$, then each of v and x cannot be a special 5_2-vertex. So each of v and x is good, thus by (R4)(R5) sends 1 to g. So $\eta^*(g) \geq -2 + 1 + 1 = 0$. If $d(u) = 2$, then each of v and x is either good or a special 5_2-vertex. If at least one of v, x is a special 5_2-vertex, then g is a I-poor face. Otherwise, both v and x are good. In either case, g gets 1 from each of v and x by (R3)-(R5). So $\eta^*(g) \geq -2 + 1 + 1 = 0$.

Case 4.2: u is a 3^--vertex and v is bad. Then $d(u) = 2$ by Corollary 2.6(1). Then $d(x) \geq 4$ by Lemma 2.2 and $d(w) \geq 4$ by Lemma 2.3 and Corollary 2.6(1). So w cannot be a special 5_2-vertex. By Corollary 2.6(1) w is good. If x is also good, then each of w and x sends 1 to f by (R4)(R5). So $\eta^*(g) \geq -2 + 1 + 1 = 0$. So we may assume that x is bad, semi-bad or a special 5_2-vertex. By Lemma 2.4(5) $d(w) \geq 5$. If w is a 5_0-vertex or a 6^+-vertex, then w sends 2 to f by (R4)(R5). So $\eta^*(g) \geq -2 + 2 = 0$. So we assume that w is a 5-vertex with at least one incident 2-vertex. By Lemma 2.4(1) x must be a special 5_2-vertex. By Lemma 2.4(5) w is incident to exactly one 2-vertex. In this case, g is a II-poor face. So g gets 1 from each of w and x by (R3)(R4). So $\eta^*(g) \geq -2 + 1 + 1 = 0$. □

Lemma 3.2. *Each vertex x in G has $\eta^*(x) \geq 0$.*

Proof. Let x be a k-vertex in G. By Lemma 2.1 $k \geq 2$. If $k = 2$, then x is incident to two 4^+-vertices. So x gets 1 from each of the two neighbors by (R1). So $\eta^*(x) \geq -2 + 1 + 1 = 0$. If $k = 3$, then x is not involved in the discharging procedure, so $\eta^*(x) = \eta(x) = 0$. So we are left to consider the following three cases.

Case 1: $k = 4$. Then x is incident to at most two 4-faces since G contains no 6-cycles and $(2, 4^+, 2, 4^+)$-faces by Lemma 2.3. By Corollary 2.6(3) x has at most two 2-neighbors. If x is a 4_0-vertex, then x sends 1 to each of at most two 4-faces by (R4). If x is a 4_1-vertex, then x sends 1 to the 2-neighbor, $\frac{1}{2}$ to each incident 4-face by (R1)(R2). If x is a 4_2-vertex, then x sends 1 to each 2-neighbor by (R1). So in any case, we have $\eta^*(x) \geq 2 \times 4 - 6 - \max\{1 + 1, 1 + \frac{1}{2} \times 2, 1 + 1\} = 0$.

Case 2: $k = 5$. Then x has at most four 2-neighbors by Corollary 2.6(3). By (R1) x sends 1 to each 2-neighbor. If x is a 5_0-vertex, then x sends 2 to each of at most two incident 4-faces by (R4). So $\eta^*(x) \geq 2 \times 5 - 6 - 2 \times 2 = 0$. If x is a 5_1-vertex, then x sends 1 to each of at most three incident 4-faces by (R4). So $\eta^*(x) \geq 2 \times 5 - 6 - 1 - 1 \times 3 = 0$. If x is a non-special 5_2-vertex, then x sends 1 to each of at most two 4-faces by (R4). So $\eta^*(x) \geq 2 \times 5 - 6 - 1 \times 2 - 1 \times 2 = 0$. If x is a special 5_2-vertex, then x is incident to at most one poor 4-face by Lemma 2.7. So x sends 1 to each incident poor 4-face and $\frac{1}{2}$ to each other incident 4-face by (R3). So $\eta^*(x) \geq 2 \times 5 - 6 - 1 \times 2 - 1 - \frac{1}{2} \times 2 = 0$. If x is a 5_3-vertex, then x is semi-bad. x sends $\frac{1}{2}$ to each of at most two incident 4-faces by (R2). So $\eta^*(x) \geq 2 \times 5 - 6 - 1 \times 3 - \frac{1}{2} \times 2 = 0$. If x is a 5_4-vertex, then x is bad. x only sends charge to each 2-neighbor. So $\eta^*(x) \geq 2 \times 5 - 6 - 1 \times 4 = 0$.

Case 3: $k \geq 6$. By (R1)(R5) x sends 1 to each 2-neighbor and each incident $(2, 6^+, 3^+, 4^+)$-face, 2 to each incident $(3^+, 6^+, 3^+, 2^+)$-face. Let $N(x) = \{x_p : 1 \leq p \leq k\}$ and g_p be the face containing x_p, x, x_{p+1} (index modulo k). Since G contains no 6-cycles and $(2, 4^+, 2, 4^+)$-faces by Lemma 2.3, x is incident to at most two consecutive $(2, 4^+, 3^+, 4^+)$-faces. Furthermore, if g_p, g_{p+1} are 4-faces, then $d(x_{p+1}) = 2$. For $1 \leq p \leq k$, if $d(g_p) = 4$, then we call g_p a non-isolated 4-face when one of g_{p-1}, g_{p+1} is also a 4-face, otherwise, we call g_p an isolated 4-face. Let $t_1(x)$ be the number of non-isolated 4-faces incident to x. Let $t_2(x)$ be the number of isolated 4-faces incident to x. Let $n_2(x)$ be the number of incident 2-vertices of x not on any 4-face. By (R1)(R5) x sends 1 to each incident 2-vertex and each incident $(2, 6^+, 3^+, 4^+)$-face, 2 to each incident $(3^+, 6^+, 3^+, 2^+)$-face. So we may regard the charge x send out as x sends 3 to each of two non-isolated 4-faces incident to x, 2 to each isolated 4-face incident to x and 1 to each incident 2-vertices of x not on any 4-face. So $\eta^*(x) \geq 2d(x) - 6 - 3t_1 - 2t_2 - n_2(x) \geq 2d(x) - d(x) - 6 = d(x) - 6 \geq 0$. □

Data availability

No data was used for the research described in the article.

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